PROPER PUSH-PULL: A_Y^{-1} F_{SMALL,*} = F_{BIG,PH,*} A_X^{-1}, AND ITS DERIVED FORM FOR TORSION COMPLEXES

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Editorial note. The counted statement is the author's own lemma environment labelled lemma-proper-push-pull-ph-etale (source lines 2059-2068); the proof environment beginning on the line after the following blank line (source line 2070) is its own proof and closes at source line 2140.

Editorial note. The proof invokes four results of the same chapter by label and one cross-chapter result (Cohomology on Sites); the four chapter results are retained as context spans with the generator's transitive expansion, and the cross-chapter result is declared as a tagged source pointer.

Editorial note. The author's proof uses the phrase 'You can prove this by repeating the proof of ... we will instead deduce the result from this'; both the remark and the alternative route are the author's own and are reproduced verbatim.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: proper push-pull for the ph topology against the etale topology, in its underived and derived forms.

Allowed standard prerequisites (editorial).

- The comparison between the ph and etale topologies. The maps a_X, epsilon_X and the identity epsilon_{X,*} o a_X^{-1} = pi_X^{-1}, plus the vanishing R^j epsilon_{X,*} = 0 for j > 0 on the relevant sheaves (the author's lemma-comparison-ph-etale, retained as context).. It rewrites a_Y^{-1} through epsilon_{Y,*} at source lines 2074-2077 and supplies the vanishing in the induction at source lines 2129-2132.
- Push-pull for the ph topology. The commutative diagram relating f_{big,ph,*} and f_{big,etale,*} (the author's lemma-push-pull-ph-etale, retained as context).. It gives the first equality in the derived chain at source lines 2107-2108.
- Proper comparison of higher direct images. The two parts of the proper comparison statement pi_Y^{-1} R f_{small,*} = R f_{big,etale,*} pi_X^{-1} (the author's lemma-compare-higher-direct-image-proper, retained as context).. Part (1) finishes the proof of (1) at source lines 2091-2092 and part (2) gives the third equality at source lines 2112-2113.

Date: stacks-project:spaces-more-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Cohomological descent for the ph topology. The identity R f_{big,etale,*} pi_X^{-1} K = pi_Y^{-1} R f_{small,*} K for the etale and ph topologies (the author's lemma-cohomological-descent-etale-ph, retained as context).. It gives the second and fourth equalities of the derived chain at source lines 2111 and 2114.

Source reference. Original lemma 0D6E: sources/stacks-chow/tags.txt, source lines 16519–16519. Cross-chapter pointer whose target carries the canonical tag 0D6E in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFF: sources/stacks-chow/tags.txt, source lines 16835–16835. Cross-chapter pointer whose target carries the canonical tag 0DFF in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 07A7: sources/stacks-chow/tags.txt, source lines 9297–9297. Cross-chapter pointer whose target carries the canonical tag 07A7 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0A4K: sources/stacks-chow/tags.txt, source lines 12778–12778. Cross-chapter pointer whose target carries the canonical tag 0A4K in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DE4: sources/stacks-chow/tags.txt, source lines 16789–16789. Cross-chapter pointer whose target carries the canonical tag 0DE4 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 072W: sources/stacks-chow/tags.txt, source lines 9040–9040. Cross-chapter pointer whose target carries the canonical tag 072W in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03F3: sources/stacks-chow/tags.txt, source lines 4478–4478. Cross-chapter pointer whose target carries the canonical tag 03F3 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDW: sources/stacks-chow/tags.txt, source lines 16781–16781. Cross-chapter pointer whose target carries the canonical tag 0DDW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DD7: sources/stacks-chow/tags.txt, source lines 16757–16757. Cross-chapter pointer whose target carries the canonical tag 0DD7 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Cross-chapter pointer whose target carries the canonical tag 015E in

the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0734: sources/stacks-chow/tags.txt, source lines 9048–9048. Cross-chapter pointer whose target carries the canonical tag 0734 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDA: sources/stacks-chow/tags.txt, source lines 16760–16760. Cross-chapter pointer whose target carries the canonical tag 0DDA in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0EZ0: sources/stacks-chow/tags.txt, source lines 18710–18710. Cross-chapter pointer whose target carries the canonical tag 0EZ0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0FN5: sources/stacks-chow/tags.txt, source lines 19555–19555. Cross-chapter pointer whose target carries the canonical tag 0FN5 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFM: sources/stacks-chow/tags.txt, source lines 16842–16842. Cross-chapter pointer whose target carries the canonical tag 0DFM in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFH: sources/stacks-chow/tags.txt, source lines 16837–16837. Cross-chapter pointer whose target carries the canonical tag 0DFH in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09XN: sources/stacks-chow/tags.txt, source lines 12536–12536. Cross-chapter pointer whose target carries the canonical tag 09XN in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1510–1510. Cross-chapter pointer whose target carries the canonical tag 015J in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 089J: sources/stacks-chow/tags.txt, source lines 10499–10499. Cross-chapter pointer whose target carries the canonical tag 089J in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0A3U: sources/stacks-chow/tags.txt, source lines 12752–12752. Cross-chapter pointer whose target carries the canonical tag 0A3U in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03CG: sources/stacks-chow/tags.txt, source lines 4383–4383. Cross-chapter pointer whose target carries the canonical tag 03CG in

the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03A0: sources/stacks-chow/tags.txt, source lines 4291–4291. Cross-chapter pointer whose target carries the canonical tag 03A0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 015Z: sources/stacks-chow/tags.txt, source lines 1526–1526. Cross-chapter pointer whose target carries the canonical tag 015Z in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 08BJ: sources/stacks-chow/tags.txt, source lines 10569–10569. Cross-chapter pointer whose target carries the canonical tag 08BJ in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 021F: sources/stacks-chow/tags.txt, source lines 2668–2668. Cross-chapter pointer whose target carries the canonical tag 021F in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04BG: sources/stacks-chow/tags.txt, source lines 5651–5651. Cross-chapter pointer whose target carries the canonical tag 04BG in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04BH: sources/stacks-chow/tags.txt, source lines 5652–5652. Cross-chapter pointer whose target carries the canonical tag 04BH in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 021I: sources/stacks-chow/tags.txt, source lines 2671–2671. Cross-chapter pointer whose target carries the canonical tag 021I in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 08FI: sources/stacks-chow/tags.txt, source lines 10711–10711. Cross-chapter pointer whose target carries the canonical tag 08FI in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0758: sources/stacks-chow/tags.txt, source lines 9122–9122. Cross-chapter pointer whose target carries the canonical tag 0758 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter pointer whose target carries the canonical tag 04K2 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DES: sources/stacks-chow/tags.txt, source lines 16812–16812. Cross-chapter pointer whose target carries the canonical tag 0DES in

the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 095S: sources/stacks-chow/tags.txt, source lines 11595–11595. Cross-chapter pointer whose target carries the canonical tag 095S in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 07U6: sources/stacks-chow/tags.txt, source lines 9961–9961. Cross-chapter pointer whose target carries the canonical tag 07U6 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Then there are commutative diagrams of topoi*

$$\begin{array}{ccc} Sh((Spaces/X)_{ph}) & \xrightarrow{f_{big,ph}} & Sh((Spaces/Y)_{ph}) \\ \downarrow{\scriptstyle \epsilon_X} & & \downarrow{\scriptstyle \epsilon_Y} \\ Sh((Spaces/X)_{étale}) & \xrightarrow{f_{big,étale}} & Sh((Spaces/Y)_{étale}) \end{array}$$

and

$$\begin{array}{ccc} Sh((Spaces/X)_{ph}) & \xrightarrow{f_{big,ph}} & Sh((Spaces/Y)_{ph}) \\ \downarrow{\scriptstyle a_X} & & \downarrow{\scriptstyle a_Y} \\ Sh(X_{étale}) & \xrightarrow{f_{small}} & Sh(Y_{étale}) \end{array}$$

with $a_X = \pi_X \circ \epsilon_X$ and $a_Y = \pi_X \circ \epsilon_X$.

Proof. This follows immediately from working out the definitions of the morphisms involved, see Topologies on Spaces, Section 0DFF and Section 1. □

Lemma 0.2. *Let S be a scheme. Let $f : Y \to X$ be a proper morphism of algebraic spaces over S. Then we have*

1. *$\pi_X^{-1} \circ f_{small,*} = f_{big,*} \circ \pi_Y^{-1}$ as functors $Sh(Y_{étale}) \to Sh((Spaces/X)_{étale})$,*
2. *$\pi_X^{-1} R f_{small,*} K = R f_{big,*} \pi_Y^{-1} K$ for K in $D^+(Y_{étale})$ whose cohomology sheaves are torsion, and*
3. *$\pi_X^{-1} R f_{small,*} K = R f_{big,*} \pi_Y^{-1} K$ for all K in $D(Y_{étale})$ if f is finite.*

Proof. Proof of (1). Let $\mathcal{F}$ be a sheaf on $Y_{étale}$. Let $g : X' \to X$ be an object of $(Spaces/X)_{étale}$. Consider the fibre product

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} \\ Y & \xrightarrow{f} & X \end{array}$$

Then we have

$$(f_{big,*}\pi_Y^{-1}\mathcal{F})(X') = (\pi_Y^{-1}\mathcal{F})(Y') = ((g'_{small})^{-1}\mathcal{F})(Y') = (f'_{small,*}(g'_{small})^{-1}\mathcal{F})(X')$$

the second equality by Lemma 1.2. On the other hand

$$(\pi_X^{-1} f_{small,*}\mathcal{F})(X') = (g_{small}^{-1} f_{small,*}\mathcal{F})(X')$$

again by Lemma 1.2. Hence by proper base change for sheaves of sets (Lemma 1.4) we conclude the two sets are canonically isomorphic. The isomorphism is compatible with restriction mappings and defines an isomorphism $\pi_X^{-1} f_{small,*}\mathcal{F} = f_{big,*}\pi_Y^{-1}\mathcal{F}$. Thus an isomorphism of functors $\pi_X^{-1} \circ f_{small,*} = f_{big,*} \circ \pi_Y^{-1}$.

Proof of (2). There is a canonical base change map $\pi_X^{-1} Rf_{small,*}K \to Rf_{big,*}\pi_Y^{-1}K$ for any K in $D(Y_{étale})$, see Cohomology on Sites, Remark 07A7. To prove it is an isomorphism, it suffices to prove the pull back of the base change map by $i_g : Sh(X'_{étale}) \to Sh((Sch/X)_{étale})$ is an isomorphism for any object $g : X' \to X$ of $(Sch/X)_{étale}$. Let T', g', f' be as in the previous paragraph. The pullback of the base change map is

$$\begin{aligned} g_{small}^{-1} Rf_{small,*}K &= i_g^{-1}\pi_X^{-1} Rf_{small,*}K \\ &\to i_g^{-1} Rf_{big,*}\pi_Y^{-1}K \\ &= Rf'_{small,*}(i_{g'}^{-1}\pi_Y^{-1}K) \\ &= Rf'_{small,*}((g'_{small})^{-1}K) \end{aligned}$$

where we have used $\pi_X \circ i_g = g_{small}$, $\pi_Y \circ i_{g'} = g'_{small}$, and Lemma 1.1. This map is an isomorphism by the proper base change theorem (Lemma 1.3) provided K is bounded below and the cohomology sheaves of K are torsion.

Proof of (3). If f is finite, then the functors $f_{small,*}$ and $f_{big,*}$ are exact. This follows from Cohomology of Spaces, Lemma 0A4K for f_{small}. Since any base change f' of f is finite too, we conclude from Lemma 1.1 part (3) that $f_{big,*}$ is exact too (as the higher derived functors are zero). Thus this case follows from part (1). □

Lemma 0.3. *Let S be a scheme. Let X be an algebraic space over S.*

(1) *For $\mathcal{F} \in Sh(X_{étale})$ we have $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$ and $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$.*
(2) *For $\mathcal{F} \in Ab(X_{étale})$ torsion we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$.*

Proof. We have $a_X^{-1}\mathcal{F} = \epsilon_X^{-1}\pi_X^{-1}\mathcal{F}$. By Lemma 1.2 the étale sheaf $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology and therefore is equal to $a_X^{-1}\mathcal{F}$ (as pulling back by ϵ_X is given by ph sheafification). Recall moreover that $\epsilon_{X,*}$ is the identity on underlying presheaves. Now part (1) is immediate from the explicit description of π_X^{-1} in Lemma 1.2.

We will prove part (2) by reducing it to the case of schemes – see part (1) of Étale Cohomology, Lemma 0DE4. This will "clearly work" as every algebraic space is étale locally a scheme. The details are given below but we urge the reader to skip the proof.

For an abelian sheaf $\mathcal{H}$ on $(Spaces/X)_{ph}$ the higher direct image $R^p\epsilon_{X,*}\mathcal{H}$ is the sheaf associated to the presheaf $U \mapsto H^p_{ph}(U, \mathcal{H})$ on $(Spaces/X)_{étale}$. See Cohomology on Sites, Lemma 072W. Since every object of $(Spaces/X)_{étale}$ has a covering by schemes, it suffices to prove that given U/X a scheme and $\xi \in H^p_{ph}(U, a_X^{-1}\mathcal{F})$ we can find an étale covering $\{U_i \to U\}$ such that ξ restricts to zero on U_i. We have

$$\begin{aligned} H^p_{ph}(U, a_X^{-1}\mathcal{F}) &= H^p((Spaces/U)_{ph}, (a_X^{-1}\mathcal{F})|_{Spaces/U}) \\ &= H^p((Sch/U)_{ph}, (a_X^{-1}\mathcal{F})|_{Sch/U}) \end{aligned}$$

where the second identification is Lemma 1.5 and the first is a general fact about restriction (Cohomology on Sites, Lemma 03F3). Looking at the first paragraph and

the corresponding result in the case of schemes (Étale Cohomology, Lemma 0DDW) we conclude that the sheaf $(a_X^{-1}\mathcal{F})|_{Sch/U}$ matches the pullback by the "schemes version of a_U". Therefore we can find an étale covering $\{U_i \to U\}$ such that our class dies in $H^p((Sch/U_i)_{ph}, (a_X^{-1}\mathcal{F})|_{Sch/U_i})$ for each i, see Étale Cohomology, Lemma 0DE4 (the precise statement one should use here is that V_n holds for all n which is the statement of part (2) for the case of schemes). Transporting back (using the same formulas as above but now for U_i) we conclude ξ restricts to zero over U_i as desired. □

Lemma 0.4. *Let S be a scheme. Let X be an algebraic space over S. For $K \in D^+(X_{étale})$ with torsion cohomology sheaves the maps*

$$\pi_X^{-1}K \longrightarrow R\epsilon_{X,*}a_X^{-1}K \quad \text{and} \quad K \longrightarrow Ra_{X,*}a_X^{-1}K$$

are isomorphisms with $a_X : Sh((Spaces/X)_{ph}) \to Sh(X_{étale})$ as above.

Proof. We only prove the second statement; the first is easier and proved in exactly the same manner. There is a reduction to the case where K is given by a single torsion abelian sheaf. Namely, represent K by a bounded below complex $\mathcal{F}^\bullet$ of torsion abelian sheaves. This is possible by Cohomology on Sites, Lemma 0DD7. By the case of a sheaf we see that $\mathcal{F}^n = a_{X,*}a_X^{-1}\mathcal{F}^n$ and that the sheaves $R^q a_{X,*}a_X^{-1}\mathcal{F}^n$ are zero for $q > 0$. By Leray's acyclicity lemma (Derived Categories, Lemma 015E) applied to $a_X^{-1}\mathcal{F}^\bullet$ and the functor $a_{X,*}$ we conclude. From now on assume $K = \mathcal{F}$ where $\mathcal{F}$ is a torsion abelian sheaf.

By Lemma 0.3 we have $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$. Thus it suffices to show that $R^q a_{X,*}a_X^{-1}\mathcal{F} = 0$ for $q > 0$. For this we can use $a_X = \epsilon_X \circ \pi_X$ and the Leray spectral sequence (Cohomology on Sites, Lemma 0734). By Lemma 0.3 we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$. We have $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$ and by Lemma 1.6 we have $R^j\pi_{X,*}(\pi_X^{-1}\mathcal{F}) = 0$ for $j > 0$. This concludes the proof. □

1. Comparing big and small topoi

Lemma 1.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S.*

1. *For K in $D((Spaces/Y)_{étale})$ we have $(Rf_{big,*}K)|_{X_{étale}} = Rf_{small,*}(K|_{Y_{étale}})$ in $D(X_{étale})$.*
2. *For K in $D((Spaces/Y)_{étale}, \mathcal{O})$ we have $(Rf_{big,*}K)|_{X_{étale}} = Rf_{small,*}(K|_{Y_{étale}})$ in $D(Mod(X_{étale}, \mathcal{O}_X))$.*

More generally, let $g : X' \to X$ be an object of $(Spaces/X)_{étale}$. Consider the fibre product

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{g} & X \end{array}$$

Then

3. *For K in $D((Spaces/Y)_{étale})$ we have $i_g^{-1}(Rf_{big,*}K) = Rf'_{small,*}(i_{g'}^{-1}K)$ in $D(X'_{étale})$.*
4. *For K in $D((Spaces/Y)_{étale}, \mathcal{O})$ we have $i_g^*(Rf_{big,*}K) = Rf'_{small,*}(i_{g'}^*K)$ in $D(Mod(X'_{étale}, \mathcal{O}_{X'}))$.*

(5) *For K in $D((Spaces/Y)_{étale})$ we have $g_{big}^{-1}(Rf_{big,*}K) = Rf'_{big,*}((g'_{big})^{-1}K)$ in $D((Spaces/X')_{étale})$.*
(6) *For K in $D((Spaces/Y)_{étale}, \mathcal{O})$ we have $g_{big}^{*}(Rf_{big,*}K) = Rf'_{big,*}((g'_{big})^{*}K)$ in $D(Mod(X'_{étale}, \mathcal{O}_{X'}))$.*

Proof. Part (1) follows from Lemma 1.7 and (1.6.1) on choosing a K-injective complex of abelian sheaves representing K.

Part (3) follows from Lemma 1.7 and Topologies, Lemma 0DDA on choosing a K-injective complex of abelian sheaves representing K.

Part (5) is Cohomology on Sites, Lemma 0EZ0.

Part (6) is Cohomology on Sites, Lemma 0FN5.

Part (2) can be proved as follows. Above we have seen that $\pi_X \circ f_{big} = f_{small} \circ \pi_Y$ as morphisms of ringed sites. Hence we obtain $R\pi_{X,*} \circ Rf_{big,*} = Rf_{small,*} \circ R\pi_{Y,*}$ by Cohomology on Sites, Lemma 0D6E. Since the restriction functors $\pi_{X,*}$ and $\pi_{Y,*}$ are exact, we conclude.

Part (4) follows from part (6) and part (2) applied to $f' : Y' \to X'$. □

Lemma 1.2. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\pi_X^{-1}\mathcal{F}$ is given by the rule*

$$(\pi_X^{-1}\mathcal{F})(Y) = \Gamma(Y_{étale}, f_{small}^{-1}\mathcal{F})$$

for $f : Y \to X$ in $(Spaces/X)_{étale}$. Moreover, $\pi_Y^{-1}\mathcal{F}$ satisfies the sheaf condition with respect to smooth, syntomic, fppf, fpqc, and ph coverings.

Proof. Since pullback is transitive and $f_{small} = \pi_X \circ i_f$ (see above) we see that $i_f^{-1}\pi_X^{-1}\mathcal{F} = f_{small}^{-1}\mathcal{F}$. This shows that π_X^{-1} has the description given in the lemma.

To prove that $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology it suffices by Topologies on Spaces, Lemma 0DFM to show that for a surjective proper morphism $V \to U$ of algebraic spaces over X we have $(\pi_X^{-1}\mathcal{F})(U)$ is the equalizer of the two maps $(\pi_X^{-1}\mathcal{F})(V) \to (\pi_X^{-1}\mathcal{F})(V \times_U V)$. This we have seen in Lemma 1.8.

The case of smooth, syntomic, fppf coverings follows from the case of ph coverings by Topologies on Spaces, Lemma 0DFH.

Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be an fpqc covering of algebraic spaces over X. Let $s_i \in (\pi_X^{-1}\mathcal{F})(U_i)$ be sections which agree over $U_i \times_U U_j$. We have to prove there exists a unique $s \in (\pi_X^{-1}\mathcal{F})(U)$ restricting to s_i over U_i. Case I: U and U_i are schemes. This case follows from Étale Cohomology, Lemma 09XN. Case II: U is a scheme. Here we choose surjective étale morphisms $T_i \to U_i$ where T_i is a scheme. Then $\mathcal{T} = \{T_i \to U\}$ is an fpqc covering by schemes and by case I the result holds for $\mathcal{T}$. We omit the verification that this implies the result for $\mathcal{U}$. Case III: general case. Let $W \to U$ be a surjective étale morphism, where W is a scheme. Then $\mathcal{W} = \{U_i \times_U W \to W\}$ is an fpqc covering (by algebraic spaces) of the scheme W. By case II the result hold for $\mathcal{W}$. We omit the verification that this implies the result for $\mathcal{U}$. □

Lemma 1.3. *Let S be a scheme. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian square of algebraic spaces over S. Assume f is proper. Let $E \in D^+(X_{étale})$ have torsion cohomology sheaves. Then the base change map $g^{-1}Rf_*E \to Rf'_*(g')^{-1}E$ is an isomorphism.

Proof. This is a simple consequence of the proper base change theorem (Theorem 1.9) using the spectral sequences

$$E_2^{p,q} = R^pf_*H^q(E) \quad \text{and} \quad {E'}_2^{p,q} = R^pf'_*(g')^{-1}H^q(E)$$

converging to R^nf_*E and $R^nf'_*(g')^{-1}E$. The spectral sequences are constructed in Derived Categories, Lemma 015J. Some details omitted. □

Lemma 1.4. *Let S be a scheme. Let $f : X \to Y$ and $g : Y' \to Y$ be a morphisms of algebraic spaces over S. Assume f is proper. Set $X' = Y' \times_Y X$ with projections $f' : X' \to Y'$ and $g' : X' \to X$. Let $\mathcal{F}$ be any sheaf on $X_{étale}$. Then $g^{-1}f_*\mathcal{F} = f'_*(g')^{-1}\mathcal{F}$.*

Proof. The question is étale local on Y'. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme V' and a surjective étale morphism $V' \to V \times_Y Y'$. Then we may replace Y' by V' and Y by V. Hence we may assume Y and Y' are schemes. Then we may work Zariski locally on Y and Y' and hence we may assume Y and Y' are affine schemes.

Assume Y and Y' are affine schemes. Choose a surjective proper morphism $h_1 : X_1 \to X$ where X_1 is a scheme, see Cohomology of Spaces, Lemma 089J. Set $X_2 = X_1 \times_X X_1$ and denote $h_2 : X_2 \to X$ the structure morphism. Observe this is a scheme. By the case of schemes (Étale Cohomology, Lemma 0A3U) we know the lemma is true for the cartesian diagrams

$$\begin{array}{ccc} X'_1 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array} \quad \text{and} \quad \begin{array}{ccc} X'_2 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

and the sheaves $\mathcal{F}_i = (X_i \to X)^{-1}\mathcal{F}$. By Lemma 1.8 we have an exact sequence $0 \to \mathcal{F} \to h_{1,*}\mathcal{F}_1 \to h_{2,*}\mathcal{F}_2$ and similarly for $(g')^{-1}\mathcal{F}$ because $X'_2 = X'_1 \times_{X'} X'_1$. Hence we conlude that the lemma is true (some details omitted). □

Lemma 1.5. *Let S be a scheme. Let $\tau \in \{étale, fppf, ph\}$ (add more here). The inclusion functor*

$$(Sch/S)_\tau \longrightarrow (Spaces/S)_\tau$$

is a special cocontinuous functor (Sites, Definition 03CG) and hence identifies topoi.

Proof. The conditions of Sites, Lemma 03A0 are immediately verified as our functor is fully faithful and as every algebraic space has an étale covering by schemes. □

Lemma 1.6. *Let S be a scheme. Let X be an algebraic space over S. For $K \in D(X_{étale})$ the map*

$$K \longrightarrow R\pi_{X,*}\pi_X^{-1}K$$

is an isomorphism where $\pi_X : Sh((Spaces/X)_{étale}) \to Sh(X_{étale})$ is as above.

Proof. This is true because both π_X^{-1} and $\pi_{X,*} = i_X^{-1}$ are exact functors and the composition $\pi_{X,*} \circ \pi_X^{-1}$ is the identity functor. □

$$(f_{big,*}\mathcal{F})|_{X_{étale}} = f_{small,*}(\mathcal{F}|_{Y_{étale}}) \tag{1.6.1}$$

Lemma 1.7. *Let S be a scheme. Let $Y \to X$ be a morphism of $(Spaces/S)_{étale}$.*

1. *If $\mathcal{I}$ is injective in $Ab((Spaces/X)_{étale})$, then*
   (a) *$i_f^{-1}\mathcal{I}$ is injective in $Ab(Y_{étale})$,*
   (b) *$\mathcal{I}|_{X_{étale}}$ is injective in $Ab(X_{étale})$,*
2. *If $\mathcal{I}^\bullet$ is a K-injective complex in $Ab((Spaces/X)_{étale})$, then*
   (a) *$i_f^{-1}\mathcal{I}^\bullet$ is a K-injective complex in $Ab(Y_{étale})$,*
   (b) *$\mathcal{I}^\bullet|_{X_{étale}}$ is a K-injective complex in $Ab(X_{étale})$,*

The corresponding statements for modules do not hold.

Proof. Parts (1)(b) and (2)(b) follow formally from the fact that the restriction functor $\pi_{X,*} = i_X^{-1}$ is a right adjoint of the exact functor π_X^{-1}, see Homology, Lemma 015Z and Derived Categories, Lemma 08BJ.

Parts (1)(a) and (2)(a) can be seen in two ways. First proof: We can use that i_f^{-1} is a right adjoint of the exact functor $i_{f,!}$. This functor is constructed in Topologies, Lemma 021F for sheaves of sets and for abelian sheaves in Modules on Sites, Lemma 04BG. It is shown in Modules on Sites, Lemma 04BH that it is exact. Second proof. We can use that $i_f = i_Y \circ f_{big}$ as is shown in Topologies, Lemma 021I. Since f_{big} is a localization, we see that pullback by it preserves injectives and K-injectives, see Cohomology on Sites, Lemmas 03F3 and 08FI. Then we apply the already proved parts (1)(b) and (2)(b) to the functor i_Y^{-1} to conclude.

To see a counter example for the case of modules we refer to Étale Cohomology, Lemma 0758. □

Lemma 1.8. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective with image the equalizer of the two maps $f_*f^{-1}\mathcal{F} \to g_*g^{-1}\mathcal{F}$ where g is the structure morphism $g : Y \times_X Y \to X$.*

Proof. For any surjective morphism $f : Y \to X$ of algebraic spaces over S, the map $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective. Namely, if $\overline{x}$ is a geometric point of X, then we choose a geometric point $\overline{y}$ of Y lying over $\overline{x}$ and we consider

$$\mathcal{F}_{\overline{x}} \to (f_*f^{-1}\mathcal{F})_{\overline{x}} \to (f^{-1}\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}}$$

See Properties of Spaces, Lemma 04K2 for the last equality.

The second statement is local on X in the étale topology, hence we may and do assume Y is an affine scheme.

Choose a surjective proper morphism $Z \to Y$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. The result for $Z \to X$ implies the result for $Y \to X$.

Since $Z \to X$ is a surjective proper morphism of schemes and hence a ph covering (Topologies, Lemma 0DES) the result for $Z \to X$ follows from Étale Cohomology, Lemma 0DDW (in fact it is in some sense equivalent to this lemma). □

Theorem 1.9. *Let S be a scheme. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian square of algebraic spaces over S. Assume f is proper. Let $\mathcal{F}$ be an abelian torsion sheaf on $X_{étale}$. Then the base change map

$$g^{-1}Rf_*\mathcal{F} \longrightarrow Rf'_*(g')^{-1}\mathcal{F}$$

is an isomorphism.

Proof. This proof repeats a few of the arguments given in the proof of the proper base change theorem for schemes. See Étale Cohomology, Section 095S for more details.

The statement is étale local on Y' and Y, hence we may assume both Y and Y' are affine schemes. Observe that this in particular proves the theorem in case f is representable (we will use this below).

For every $n \geq 1$ let $\mathcal{F}[n]$ be the subsheaf of sections of $\mathcal{F}$ annihilated by n. Then $\mathcal{F} = \operatorname{colim} \mathcal{F}[n]$. By Cohomology of Spaces, Lemma 07U6 the functors $g^{-1}R^pf_*$ and $R^pf'_*(g')^{-1}$ commute with filtered colimits. Hence it suffices to prove the theorem if $\mathcal{F}$ is killed by n.

Let $\mathcal{F} \to \mathcal{I}^\bullet$ be a resolution by injective sheaves of $\mathbf{Z}/n\mathbf{Z}$-modules. Observe that $g^{-1}f_*\mathcal{I}^\bullet = f'_*(g')^{-1}\mathcal{I}^\bullet$ by Lemma 1.4. Applying Leray's acyclicity lemma (Derived Categories, Lemma 015E) we conclude it suffices to prove $R^pf'_*(g')^{-1}\mathcal{I}^m = 0$ for $p > 0$ and $m \in \mathbf{Z}$.

Choose a surjective proper morphism $h : Z \to X$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. Choose an injective map $h^{-1}\mathcal{I}^m \to \mathcal{J}$ where $\mathcal{J}$ is an injective sheaf of $\mathbf{Z}/n\mathbf{Z}$-modules on $Z_{étale}$. Since h is surjective the map $\mathcal{I}^m \to h_*\mathcal{J}$ is injective (see Lemma 1.8). Since $\mathcal{I}^m$ is injective we see that $\mathcal{I}^m$ is a direct summand of $h_*\mathcal{J}$. Thus it suffices to prove the desired vanishing for $h_*\mathcal{J}$.

Denote h' the base change by g and denote $g'' : Z' \to Z$ the projection. There is a spectral sequence

$$E_2^{p,q} = R^pf'_*R^qh'_*(g'')^{-1}\mathcal{J}$$

converging to $R^{p+q}(f' \circ h')_*(g'')^{-1}\mathcal{J}$. Since h and $f \circ h$ are representable (by schemes) we know the result we want holds for them. Thus in the spectral sequence we see that $E_2^{p,q} = 0$ for $q > 0$ and $R^{p+q}(f' \circ h')_*(g'')^{-1}\mathcal{J} = 0$ for $p + q > 0$. It follows that $E_2^{p,0} = 0$ for $p > 0$. Now

$$E_2^{p,0} = R^pf'_*h'_*(g'')^{-1}\mathcal{J} = R^pf'_*(g')^{-1}h_*\mathcal{J}$$

by Lemma 1.4. This finishes the proof. □

Lemma 1.10. *In Lemma 0.1 if f is proper, then we have*

(1) $a_Y^{-1} \circ f_{small,*} = f_{big,ph,*} \circ a_X^{-1}$, *and*

(2) $a_Y^{-1}(Rf_{small,*}K) = Rf_{big,ph,*}(a_X^{-1}K)$ *for* K *in* $D^+(X_{étale})$ *with torsion cohomology sheaves.*

Proof. Proof of (1). You can prove this by repeating the proof of Lemma 0.2 part (1); we will instead deduce the result from this. As $\epsilon_{Y,*}$ is the identity functor on underlying presheaves, it reflects isomorphisms. Lemma 0.3 shows that $\epsilon_{Y,*} \circ a_Y^{-1} = \pi_Y^{-1}$ and similarly for X. To show that the canonical map $a_Y^{-1} f_{small,*}\mathcal{F} \to f_{big,ph,*}a_X^{-1}\mathcal{F}$ is an isomorphism, it suffices to show that

$$\begin{aligned}
\pi_Y^{-1} f_{small,*}\mathcal{F} &= \epsilon_{Y,*}a_Y^{-1} f_{small,*}\mathcal{F} \\
&\to \epsilon_{Y,*} f_{big,ph,*}a_X^{-1}\mathcal{F} \\
&= f_{big,étale,*}\epsilon_{X,*}a_X^{-1}\mathcal{F} \\
&= f_{big,étale,*}\pi_X^{-1}\mathcal{F}
\end{aligned}$$

is an isomorphism. This is part (1) of Lemma 0.2.

To see (2) we use that

$$\begin{aligned}
R\epsilon_{Y,*}Rf_{big,ph,*}a_X^{-1}K &= Rf_{big,étale,*}R\epsilon_{X,*}a_X^{-1}K \\
&= Rf_{big,étale,*}\pi_X^{-1}K \\
&= \pi_Y^{-1}Rf_{small,*}K \\
&= R\epsilon_{Y,*}a_Y^{-1}Rf_{small,*}K
\end{aligned}$$

The first equality by the commutative diagram in Lemma 0.1 and Cohomology on Sites, Lemma 0D6E. Then second equality is Lemma 0.4. The third is Lemma 0.2 part (2). The fourth is Lemma 0.4 again. Thus the base change map $a_Y^{-1}(Rf_{small,*}K) \to Rf_{big,ph,*}(a_X^{-1}K)$ induces an isomorphism

$$R\epsilon_{Y,*}a_Y^{-1}Rf_{small,*}K \to R\epsilon_{Y,*}Rf_{big,ph,*}a_X^{-1}K$$

The proof is finished by the following remark: consider a map $\alpha : a_Y^{-1}L \to M$ with L in $D^+(Y_{étale})$ having torsion cohomology sheaves and M in $D^+((Spaces/Y)_{ph})$. If $R\epsilon_{Y,*}\alpha$ is an isomorphism, then α is an isomorphism. Namely, we show by induction on i that $H^i(\alpha)$ is an isomorphism. This is true for all sufficiently small i. If it holds for $i \leq i_0$, then we see that $R^j\epsilon_{Y,*}H^i(M) = 0$ for $j > 0$ and $i \leq i_0$ by Lemma 0.3 because $H^i(M) = a_Y^{-1}H^i(L)$ in this range. Hence $\epsilon_{Y,*}H^{i_0+1}(M) = H^{i_0+1}(R\epsilon_{Y,*}M)$ by a spectral sequence argument. Thus $\epsilon_{Y,*}H^{i_0+1}(M) = \pi_Y^{-1}H^{i_0+1}(L) = \epsilon_{Y,*}a_Y^{-1}H^{i_0+1}(L)$. This implies $H^{i_0+1}(\alpha)$ is an isomorphism (because $\epsilon_{Y,*}$ reflects isomorphisms as it is the identity on underlying presheaves) as desired. □